

吕康禾

Kanghe Lyu

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Research Interests

Representation theory; Noncommutative geometry; Mathematical physics; Algebraic geometry.

Education

Lijiang No. 1 Senior High School , Class 450 and 445 High School Diploma	Lijiang, China 09/2019 – 07/2022
Sichuan University , School of Mathematics B.Sc. Special Program in Pure Mathematics / 基地班	Chengdu, China 09/2022 – 07/2026
Southern University of Science and Technology , ICM Ph.D. in Mathematics Adviser: Vyacheslav Futorny Dissertation: _____	Shenzhen, China 09/2026 – Present

Research and Project Experience

Optimization Design of a Heliostat Field Based on Nonlinear Programming China Undergraduate Mathematical Contest in Modeling, <i>National Second Prize</i> Role: Team Leader	2023
Lei Zhenzi: A Bridge-Linked Anomaly Detection and Tracing System for Multi-Source Heterogeneous Data in Smart Grids Undergraduate Innovation and Entrepreneurship Training Program, <i>National Level</i> Role: Core Member	2024

Awards and Honors

• Sichuan University Comprehensive First-Class Scholarship, Top 5%	2023–2024
• Outstanding Undergraduate Student, Sichuan University	2023–2024
• Outstanding Student Cadre, Sichuan University	2022–2023
• Sichuan Provincial Comprehensive Quality Class A Certificate	12/2024
• First Prize, Sichuan University “Youth in Motion” Oriented Relay Competition	2024
• Outstanding Graduate of Sichuan University	10/2025

Experience

• Summer School at New Uzbekistan University Speakers: Pavel Etingof and Yuri Berest.	01/07/2024 – 15/07/2024
• International Mathematics Summer School at Westlake University	06/07/2025 – 16/07/2025
• Visiting Program at New Uzbekistan University	12/11/2025 – 23/01/2026
• Visiting Program at Southern University of Science and Technology	12/03/2026 – 12/05/2026
• Summer School for Outstanding Mathematics Students, Sichuan University	05/07/2026 – 19/07/2026
• Visiting Program at Institute for Problems of Information Transmission, Moscow	

Seminars

- **Highest Weight Representations of Infinite-Dimensional Lie Algebras** 11/2025 – 02/2026
Seminar organizer and speaker. The seminar focused on highest weight representations of infinite-dimensional Lie algebras, based on materials from a course taught by Li Ren at Sichuan University.
- **Algebraic Groups and Geometric Invariant Theory** 11/2025 – 02/2026
Seminar organizer and speaker. The seminar covered algebraic groups and related topics, with emphasis on GIT quotient.

Notes and Surveys

- **Slodowy Slices and Their Quantization**
Notes on Slodowy slices, Hamiltonian reduction, and BRST cohomology.
- **A Note on Algebraic Groups and Related Topics**
An ongoing note on algebraic groups and related structures.
- **A Note on Lie Algebras**
A hand-written Chinese note taken while learning from Zhexian Wan's and Humphreys's books.
- **A Note on Quiver Varieties, Slodowy Slices, and Affine Grassmannians**
An ongoing note related to geometric representation theory.